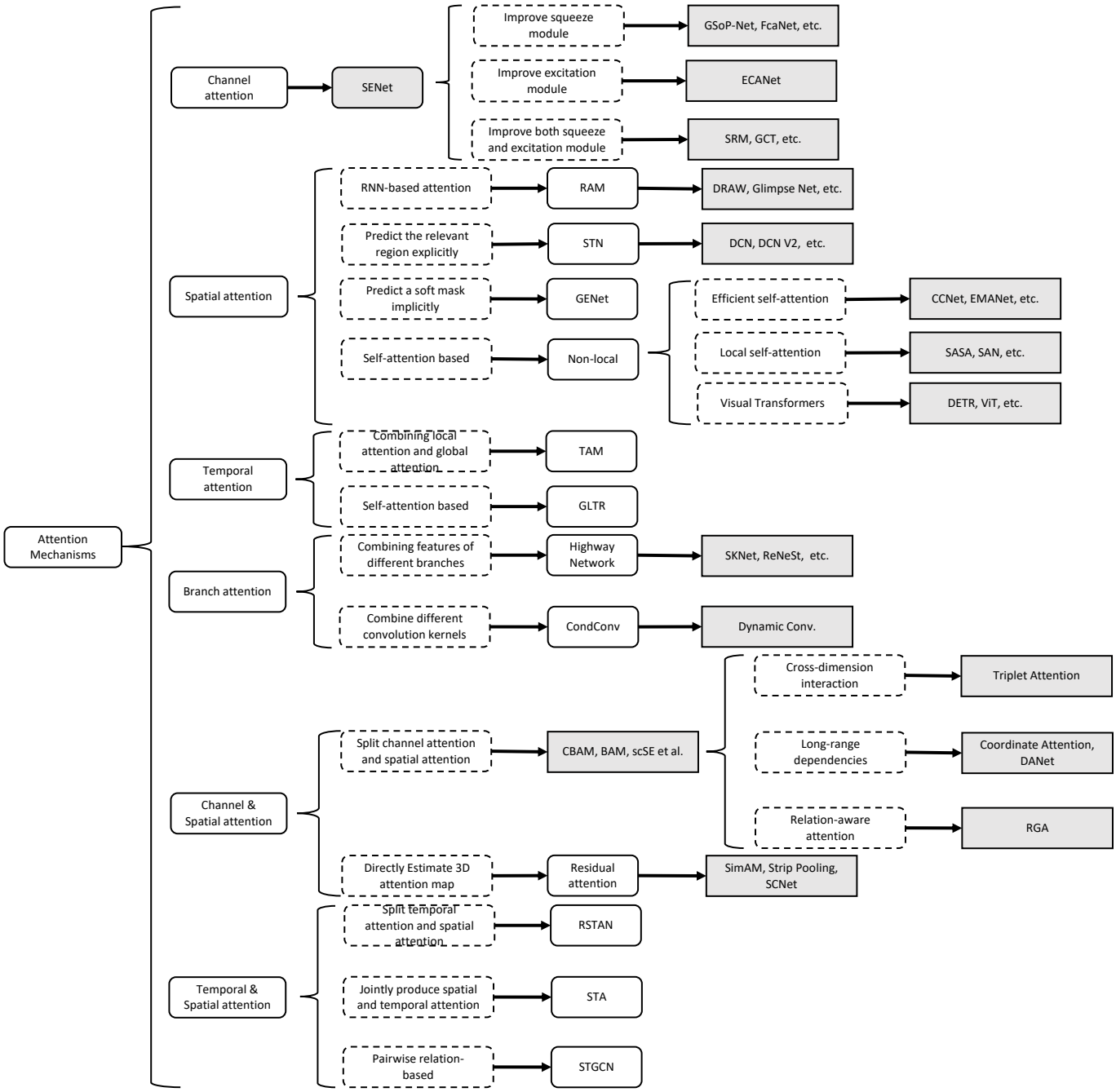

Fig. 4. Developmental context of visual attention.

excitation(SE) attention [5] as examples. For self-attention, $g(x)$ and $f(g(x), x)$ can be written as

$$Q, K, V = \text{Linear}(x) \tag{2}$$
$$g(x) = \text{Softmax}(QK) \tag{3}$$
$$f(g(x), x) = g(x)V \tag{4}$$
$$\tag{5}$$

For SE, $g(x)$ and $f(g(x), x)$ can be written as

$$g(x) = \text{Sigmoid}(\text{MLP}(\text{GAP}(x))) \tag{6}$$
$$f(g(x), x) = g(x)x \tag{7}$$

In the following, we will introduce various attention mechanisms and specify them to the above formulation.

3.2 Channel Attention

In deep neural networks, different channels in different feature maps usually represent different objects [50]. Channel attention adaptively recalibrates the weight of each channel, and can be viewed as an object selection process, thus determining *what to pay attention to*. Hu et al. [5] first proposed the concept of channel attention and presented SENet for this purpose. As Fig. 4 shows, and we discuss

shortly, three streams of work continue to improve channel attention in different ways.

In this section, we first summarize the representative channel attention works and specify process $g(x)$ and $f(g(x), x)$ described as Eq. 1 in Tab. 3 and Fig. 5. Then we discuss various channel attention methods along with their development process respectively.

3.2.1 SENet

SENet [5] pioneered channel attention. The core of SENet is a *squeeze-and-excitation* (SE) block which is used to collect global information, capture channel-wise relationships and improve representation ability.

SE blocks are divided into two parts, a squeeze module and an excitation module. Global spatial information is collected in the squeeze module by global average pooling. The excitation module captures channel-wise relationships and outputs an attention vector by using fully-connected layers and non-linear layers (ReLU and sigmoid). Then, each channel of the input feature is scaled by multiplying the corresponding element in the attention vector. Overall, a squeeze-and-excitation block F_{se} (with parameter θ) which takes X as input and outputs Y can be formulated as:

$$s = F_{\text{se}}(X, \theta) = \sigma(W_2 \delta(W_1 \text{GAP}(X))) \tag{8}$$

$$Y = sX \tag{9}$$

SE blocks play the role of emphasizing important channels while suppressing noise. An SE block can be added after each residual unit [145] due to their low computational resource requirements. However, SE blocks have shortcomings. In the squeeze module, global average pooling is too simple to capture complex global information. In the excitation module, fully-connected layers increase the complexity of the model. As Fig. 4 indicates, later works attempt to improve the outputs of the squeeze module (e.g. GSoP-Net [54]), reduce the complexity of the model by improving the excitation module (e.g. ECANet [37]), or improve both the squeeze module and the excitation module (e.g. SRM [55]).

3.2.2 GSoP-Net

An SE block captures global information by only using global average pooling (i.e. first-order statistics), which limits its modeling capability, in particular the ability to capture high-order statistics.

To address this issue, Gao et al. [54] proposed to improve the squeeze module by using a *global second-order pooling* (GSoP) block to model high-order statistics while gathering global information.

Like an SE block, a GSoP block also has a squeeze module and an excitation module. In the squeeze module, a GSoP block firstly reduces the number of channels from c to c' ($c' < c$) using a $1x1$ convolution, then computes a $c' \times c'$ covariance matrix for the different channels to obtain their correlation. Next, row-wise normalization is performed on the covariance matrix. Each (i, j) in the normalized covariance matrix explicitly relates channel i to channel j.

In the excitation module, a GSoP block performs row-wise convolution to maintain structural information and output a vector. Then a fully-connected layer and a sigmoid function are applied to get a c-dimensional attention vector. Finally, it multiplies the input features by the attention vector, as in an SE block. A GSoP block can be formulated as:

$$s = F_{\text{gsop}}(X, \theta) = \sigma(W\text{RC}(\text{Cov}(\text{Conv}(X)))) \tag{10}$$

$$Y = sX \tag{11}$$

Here, $\text{Conv}(\cdot)$ reduces the number of channels, $\text{Cov}(\cdot)$ computes the covariance matrix and $\text{RC}(\cdot)$ means row-wise convolution.

By using second-order pooling, GSoP blocks have improved the ability to collect global information over the SE block. However, this comes at the cost of additional computation. Thus, a single GSoP block is typically added after several residual blocks.

3.2.3 SRM

Motivated by successes in style transfer, Lee et al. [55] proposed the lightweight *style-based recalibration module* (SRM). SRM combines style transfer with an attention mechanism. Its main contribution is style pooling which utilizes both mean and standard deviation of the input features to improve its capability to capture global information. It also adopts a lightweight *channel-wise fully-connected* (CFC) layer, in place of the original fully-connected layer, to reduce the computational requirements.

Given an input feature map $X \in \mathbb{R}^{C \times H \times W}$, SRM first collects global information by using style pooling ($\text{SP}(\cdot)$) which combines global average pooling and global standard deviation pooling. Then a channel-wise fully connected ($\text{CFC}(\cdot)$) layer (i.e. fully connected per channel), batch normalization BN and sigmoid function σ are used to provide the attention vector. Finally, as in an SE block, the input features are multiplied by the attention vector. Overall, an SRM can be written as:

$$s = F_{\text{srm}}(X, \theta) = \sigma(\text{BN}(\text{CFC}(\text{SP}(X)))) \tag{12}$$

$$Y = sX \tag{13}$$

The SRM block improves both squeeze and excitation modules, yet can be added after each residual unit like an SE block.

3.2.4 GCT

Due to the computational demand and number of parameters of the fully connected layer in the excitation module, it is impractical to use an SE block after each convolution layer. Furthermore, using fully connected layers to model channel relationships is an implicit procedure. To overcome the above problems, Yang et al. [56] propose the *gated channel transformation* (GCT) to efficiently collect information while explicitly modeling channel-wise relationships.

Unlike previous methods, GCT first collects global information by computing the l_2-norm of each channel. Next, a learnable vector α is applied to scale the feature. Then a competition mechanism is adopted by channel normalization to interact between channels. Like other common normalization methods, a learnable scale parameter γ and bias β are applied to rescale the normalization. However, unlike previous methods, GCT adopts tanh activation to control the attention vector. Finally, it not only multiplies the input

by the attention vector but also adds an identity connection. GCT can be written as:

$$s = F_{\text{gct}}(X, \theta) = \tanh(\gamma CN(\alpha \text{Norm}(X)) + \beta) \quad (14)$$

$$Y = sX + X, \quad (15)$$

where α, β and γ are trainable parameters. Norm$(\cdot)$ indicates the $L2$-norm of each channel. CN is channel normalization.

A GCT block has fewer parameters than an SE block, and as it is lightweight, can be added after each convolutional layer of a CNN.

3.2.5 ECANet

To avoid high model complexity, SENet reduces the number of channels. However, this strategy fails to directly model correspondence between weight vectors and inputs, reducing the quality of results. To overcome this drawback, Wang et al. [37] proposed the *efficient channel attention* (ECA) block which instead uses a 1D convolution to determine the interaction between channels, instead of dimensionality reduction.

An ECA block has similar formulation to an SE block including a squeeze module for aggregating global spatial information and an efficient excitation module for modeling cross-channel interaction. Instead of indirect correspondence, an ECA block only considers direct interaction between each channel and its k-nearest neighbors to control model complexity. Overall, the formulation of an ECA block is:

$$s = F_{\text{eca}}(X, \theta) = \sigma(\text{Conv1D}(\text{GAP}(X))) \quad (16)$$

$$Y = sX \quad (17)$$

where Conv1D$(\cdot)$ denotes 1D convolution with a kernel of shape k across the channel domain, to model local cross-channel interaction. The parameter k decides the coverage of interaction, and in ECA the kernel size k is adaptively determined from the channel dimensionality C instead of by manual tuning, using cross-validation:

$$k = \psi(C) = \left| \frac{\log_2(C)}{\gamma} + \frac{b}{\gamma} \right|_{\text{odd}} \quad (18)$$

where γ and b are hyperparameters. $|x|_{\text{odd}}$ indicates the nearest odd function of x.

Compared to SENet, ECANet has an improved excitation module, and provides an efficient and effective block which can readily be incorporated into various CNNs.

3.2.6 FcaNet

Only using global average pooling in the squeeze module limits representational ability. To obtain a more powerful representation ability, Qin et al. [57] rethought global information captured from the viewpoint of compression and analysed global average pooling in the frequency domain. They proved that global average pooling is a special case of the discrete cosine transform (DCT) and used this observation to propose a novel *multi-spectral channel attention*.

Given an input feature map $X \in \mathbb{R}^{C \times H \times W}$, multi-spectral channel attention first splits X into many parts $x^i \in \mathbb{R}^{C' \times H \times W}$. Then it applies a 2D DCT to each part x^i. Note that a 2D DCT can use pre-processing results to reduce computation. After processing each part, all results are concatenated into a vector. Finally, fully connected layers, ReLU activation and a sigmoid are used to get the attention vector as in an SE block. This can be formulated as:

$$s = F_{\text{fca}}(X, \theta) = \sigma(W_2 \delta(W_1[(\text{DCT}(\text{Group}(X)))])) \quad (19)$$

$$Y = sX \quad (20)$$

where Group$(\cdot)$ indicates dividing the input into many groups and DCT$(\cdot)$ is the 2D discrete cosine transform.

This work based on information compression and discrete cosine transforms achieves excellent performance on the classification task.

3.2.7 EncNet

Inspired by SENet, Zhang et al. [53] proposed the *context encoding module* (CEM) incorporating *semantic encoding loss* (SE-loss) to model the relationship between scene context and the probabilities of object categories, thus utilizing global scene contextual information for semantic segmentation.

Given an input feature map $X \in \mathbb{R}^{C \times H \times W}$, a CEM first learns K cluster centers $D = \{d_1, \ldots, d_K\}$ and a set of smoothing factors $S = \{s_1, \ldots, s_K\}$ in the training phase. Next, it sums the difference between the local descriptors in the input and the corresponding cluster centers using soft-assignment weights to obtain a permutation-invariant descriptor. Then, it applies aggregation to the descriptors of the K cluster centers instead of concatenation for computational efficiency. Formally, CEM can be written as:

$$e_k = \frac{\sum_{i=1}^{N} e^{-s_k ||X_i - d_k||^2} (X_i - d_k)}{\sum_{j=1}^{K} e^{-s_j ||X_i - d_j||^2}} \quad (21)$$

$$e = \sum_{k=1}^{K} \phi(e_k) \quad (22)$$

$$s = \sigma(We) \quad (23)$$

$$Y = sX \quad (24)$$

where $d_k \in \mathbb{R}^C$ and $s_k \in \mathbb{R}$ are learnable parameters. ϕ denotes batch normalization with ReLU activation. In addition to channel-wise scaling vectors, the compact contextual descriptor e is also applied to compute the SE-loss to regularize training, which improves the segmentation of small objects.

Not only does CEM enhance class-dependent feature maps, but it also forces the network to consider big and small objects equally by incorporating SE-loss. Due to its lightweight architecture, CEM can be applied to various backbones with only low computational overhead.

3.2.8 Bilinear Attention

Following GSoP-Net [54], Fang et al. [146] claimed that previous attention models only use first-order information and disregard higher-order statistical information. They thus proposed a new *bilinear attention block* (bi-attention) to capture local pairwise feature interactions within each channel, while preserving spatial information.

Bi-attention employs the *attention-in-attention* (AiA) mechanism to capture second-order statistical information: the outer point-wise channel attention vectors are computed from the output of the inner channel attention. Formally,